ORF309/MAT380
Continuous Random Variables I: Expectations & Distributions

(pp. 63-70)

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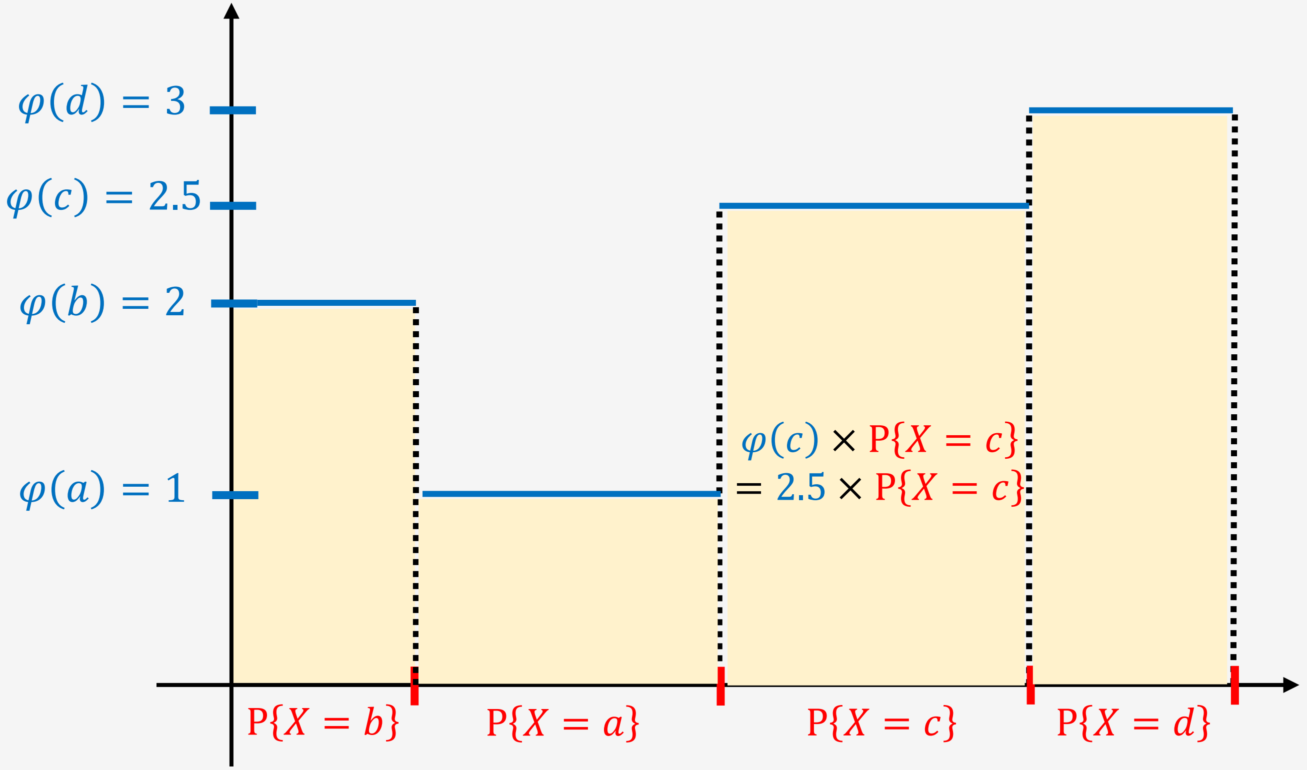
Agenda for today

- Continuous RVs
- Cumulative Distribution Function (CDF)
- Probability Distribution Function (PDF)
- Expectations
- Variances
- Example: Exponential RV

Expectations of Discrete RVs

- Let's consider the **DRV** $X: \Omega \to D$ where
  - where $D = \{a, b, c, d\}$
  - and a value function $\varphi: D \to R$
    - For example, we may have:
      - $\varphi(a) = 1,\ \varphi(b) = 2,$
      - $\varphi(c) = 2.5,\ \varphi(d) = 3$
- We also know the distribution of X
  - In general: $P\{X = x\} \neq \mathrm{P}\{X = y\}\ \forall\, x \neq y$

$$\boldsymbol{E[\varphi(X)] = \textstyle\sum_{x \in D} \varphi(x) P\{X = x\}}$$

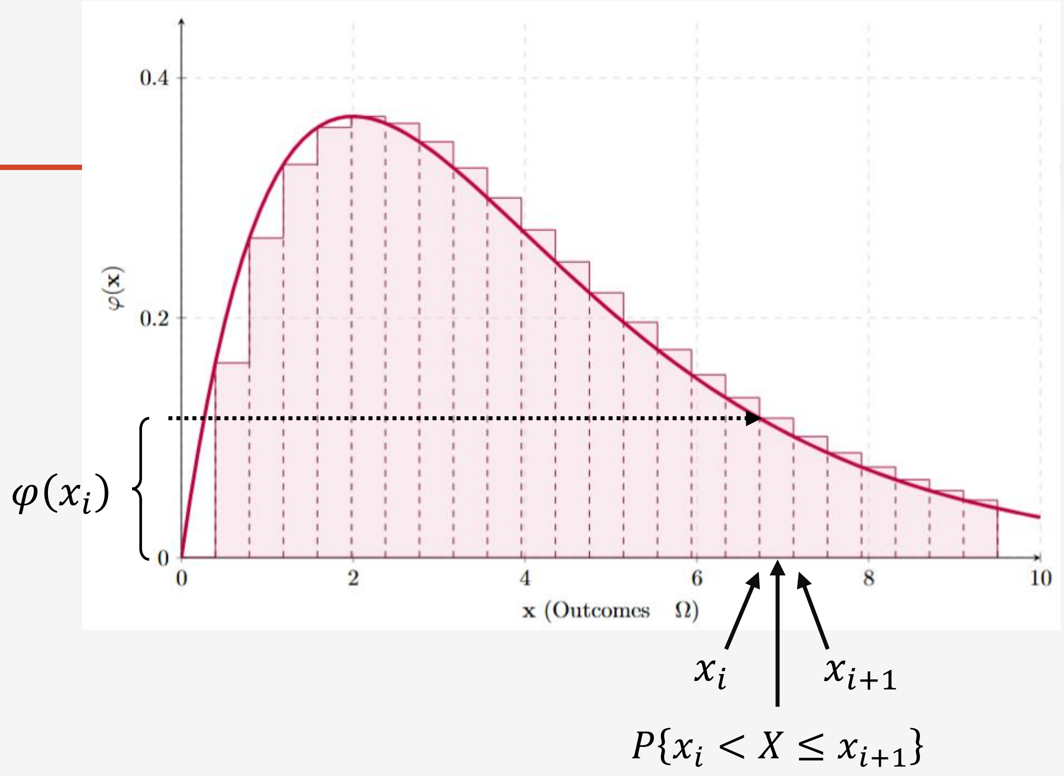


$\boldsymbol{E[\varphi(X)]}$ $\coloneqq$ **the total area under** $\boldsymbol{\varphi(x)}$

Expectations of Continuous RVs

- Let's consider the **Continuous RV** $X: \Omega \to \boldsymbol{R}$
- We cannot use the expectation formula for DRVs
  - Because $P\{X = x\} = 0$ for CRVs!!
- But the expectation should **also** be described by the **area under the value function** $\varphi(X)$
- In **Calculus** we used **integrals** to define the area under a continuous function

Expectations of Continuous RVs



$$E[\varphi(X)] = \lim_{|x_{i+1}-x_i|\to 0} \sum_i \varphi(x_i) P\{x_i < X \le x_{i+1}\} := \int \varphi(x) P\{X \in dx\}$$

You should think of $\boldsymbol{dx}$ as an infinitely small interval: $\boldsymbol{dx = (x, x + \Delta x]}$ where $\boldsymbol{\Delta x \to 0}$

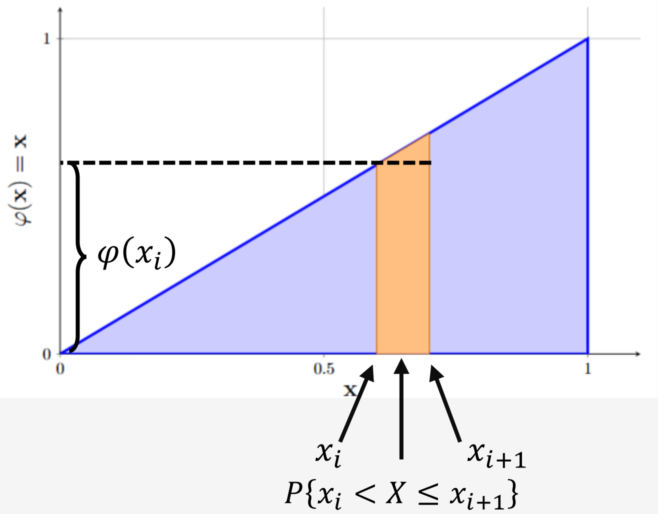
Notation: $\boldsymbol{P\{X \in dx\} := P\{x < X \le x + \Delta x\}}$ means the CRV X takes values in the small interval of length Δx

Uniform distribution for CRVs: $X \sim Unif([0,1])$

- **Example**:
  - X is a CRV that is **equally likely** to take any value in $[0,1]$
  - If $A \subseteq [0,1]$ then $\boldsymbol{P}\{\boldsymbol{A}\} = \mathbf{length}(\mathbf{A})$
  - We say that X has the **Uniform distribution** in $[0,1]$
    - **Notation**: $X \sim Unif([0,1])$
  - It is the simplest distribution
  - Let's find the **expectation of $\boldsymbol{\varphi}(\boldsymbol{X})$** for some function φ

Expectation of $X \sim Unif([0,1])$

$$E[\varphi(X)] = \lim_{|x_{i+1}-x_i| \to 0} \sum_i \varphi(x_i) \overbrace{P\{x_i < X \le x_{i+1}\}}^{\textbf{length}([x_i, x_{i+1}])} = \lim_{|x_{i+1}-x_i| \to 0} \sum_i \varphi(x_i) \overbrace{(x_{i+1} - x_i)}^{\textbf{length}([x_i, x_{i+1}])} := \int_0^1 \varphi(x)dx$$



If $\boldsymbol{\varphi(x) = x}$ we obtain the **expectation** of the CRV X:

$$E[X] = \int_0^1 x dx = \frac{1}{2} x^2 \Big|_{x=0}^{1} = \frac{1}{2}$$

This is *expected*! If every number in $[0,1]$ is equally likely, then the average should be **½**

Calculus here was very easy! It is not always the case…

Continuous RVs and Calculus

Let's try to **build bridges among fundamental concepts of Probability Theory and Calculus**

Define the function: $\boldsymbol{F(x) = P\{X \leq x\}}$ for any $CRV \quad X: \Omega \to R$

Then we compute the **limit**:

$$\lim_{|x_{i+1} - x_i| \to 0} \frac{P\{x_i < X \leq x_{i+1}\}}{x_{i+1} - x_i} =$$

average probability per unit length within the small interval

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Cumulative Distribution Function (CDF) and Probability Density Function (PDF)

For both DRVs and CRVs

- **Definition**: For a random variable X the function $\mathrm{F}(x) = P\{X \leq x\}$ is called the Cumulative Distribution Function (CDF) of X.

Only for CRVs

- **Definition**: For a random variable X the function $f(x) = \frac{dF(x)}{dx}$ is called the Probability Density Function (PDF) or simply density of X.

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- **Remark**: The PDF gives us the **instantaneous rate of change** of the probability of a CRV X at x

- **Remark**: the **distribution** of a random variable is **fully defined** by its CDF or its PDF

- **Remark**: as long as we know the **distribution** of a random variable, we can compute its **expectation**.

Properties of CDF and Density

- **Property 1**: the **CDF** is an increasing function ($F\uparrow$)
  - If $x_1 \leq x_2$ then $\{X \leq x_1\} \subseteq \{X \leq x_2\}$ which also implies

    $P\{X \leq x_1\} \leq P\{X < x_2\} \quad \Rightarrow \quad F(x_1) \leq F(x_2)$
- **Property 2**: $F(-\infty) = P\{X < -\infty\} = 0$
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Any idea how the CDF should look like?

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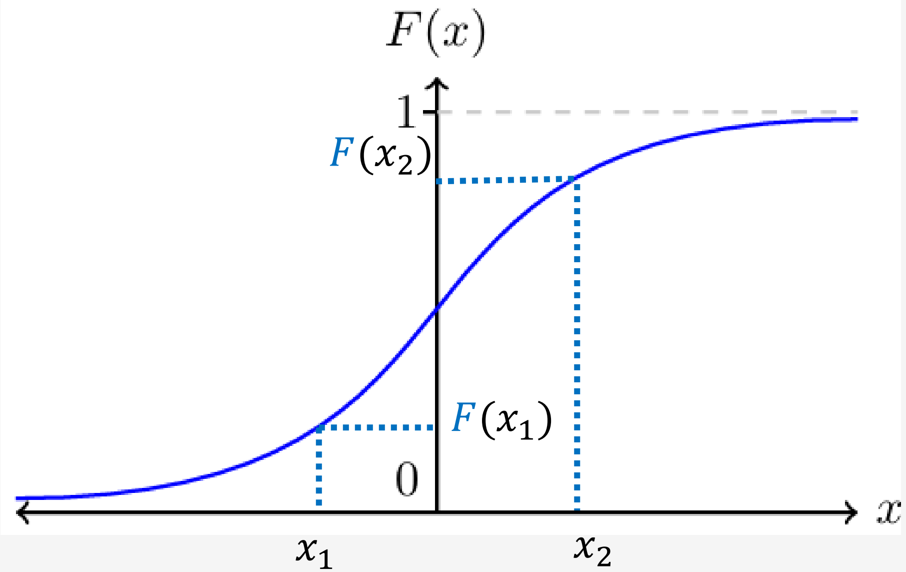

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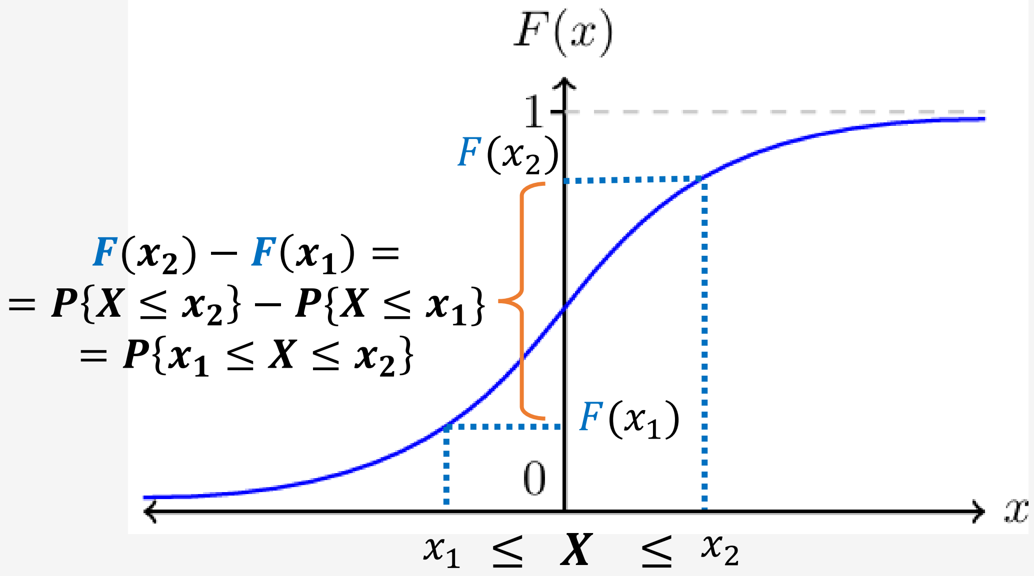


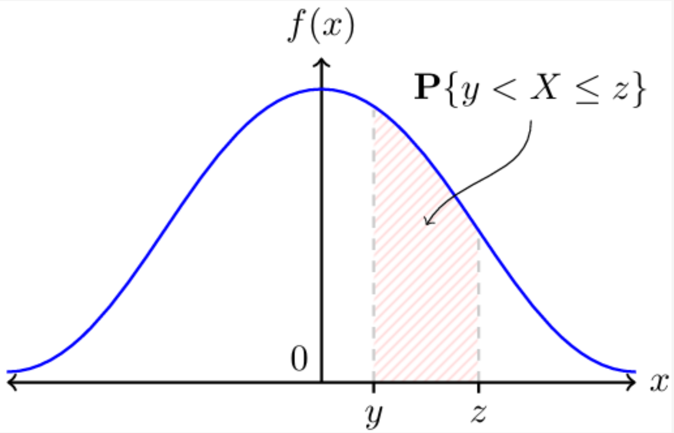
Properties of CDF and Density

- **Property 4**: the density $f(x) = \frac{dF(x)}{dx} \geq 0, \forall x$ because F is increasing
- **CLAIM**: The integral of f in $(-\infty, \infty)$ is 1 (use the definition of the Expectation)

$$1 = E[1] = \int_{-\infty}^{\infty} 1 f(x) dx = \int_{-\infty}^{\infty} f(x) dx$$

- **Probability of an interval:**

$$P\{y < X \leq z\} = E\left[\mathbf{1}_{(y,z]}(X)\right] = \int_{-\infty}^{\infty} \mathbf{1}_{(y,z]}(X) f(x) dx = \int_{y}^{z} f(x) dx$$



Properties of CDF and Density

- **Remark**: X takes values with higher probability in intervals where the **area under** the PDF is **larger**.

- **Remark**: think of PDF as the **probability** that the **value** x of the CRV X is in a small **interval** $(x, x+dx]$. That probability is: $P\{X \in dx\} = f(x)dx$

speed

distance covered

- **Remark**: Do **NOT** write $P\{X = x\}$ for CRVs!!

Exponential RVs

- **Example**:
  - Consider $T \sim Expon(\mu)$ to be **the time of the arrival of the 1st customer**
    - μ is the average **#** of arrivals per 1 hour (i.e., the rate of arrivals in 1 hour)
  - **CDF**: $F(t) = P\{T \leq t\} = 1 - e^{-\mu t}$ for $t \geq 0$
    - For $t < 0$ the CDF is defined as : $F(t) = P\{T < 0\} = 0$
  - What is the **PDF** of T ?

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$$f(t) = \frac{dF(t)}{dt} = \frac{d}{dt}(1 - e^{-\mu t}) = \mu e^{-\mu t}, \qquad t \ge 0$$

Exponential RVs

- **Example (cont.)**:
  - What is the **<u>Expectation</u>** of T ?

$$E[T] = \int_{\mathbf{0}}^{\infty} t \frac{dF(t)}{dt} dt = \int_{\mathbf{0}}^{\infty} t f(t)\, dt = \int_{0}^{\infty} t\mu e^{-\mu t}\, dt = \mu \int_{0}^{\infty} t e^{-\mu t}\, dt = ??$$

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Leibniz rule

$$= -\mu \int_{0}^{\infty} \frac{d}{d\mu}\left(e^{-\mu t}\right) dt = -\mu \frac{d}{d\mu}\left(\int_{0}^{\infty} e^{-\mu t}\, dt\right) = -\mu \frac{d}{d\mu}\left(\frac{1}{\mu}\right) = \frac{1}{\mu}$$

Exponential RVs

- **Example (cont.)**:

  - What is the **<u>Variance</u>** of $T \sim Expon(\mu)$?

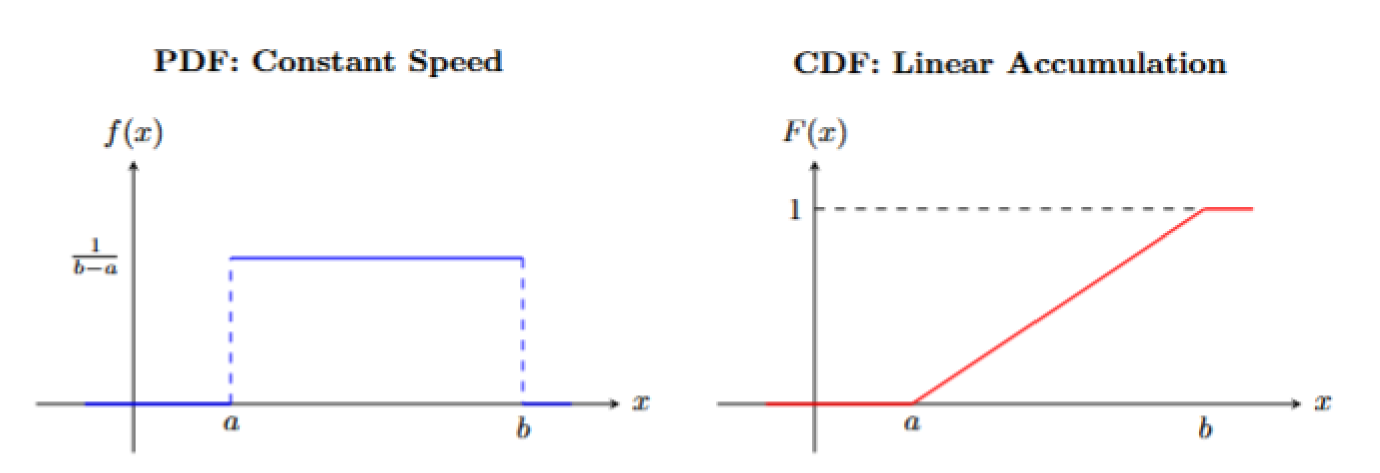
Leibniz rule

$$E[T^2] = \int_0^\infty t^2 \mu e^{-\mu t}\, dt = \mu \int_0^\infty \frac{d^2}{d\mu^2}(e^{-\mu t})\, dt = \mu \frac{d^2}{d\mu^2}\left(\int_0^\infty e^{-\mu t}\, dt\right) = -\mu \frac{d^2}{d\mu^2}\left(\frac{1}{\mu}\right) = \frac{2}{\mu^2}$$

  - Hence:

$$Var(T) = E[T^2] - (E[T])^2 = \frac{2}{\mu^2} - \frac{1}{\mu^2} = \frac{1}{\mu^2}$$

Some commonly used CRVs and their PDF and CDF



Uniform distribution $X \sim Unif([a,b])$

$$f(x) = \begin{cases} \frac{1}{b-a}, & x \in (a,b) \\ 0 & , x \notin [a,b] \end{cases} \qquad F(x) = \begin{cases} 0 & , x < a \\ \frac{x-a}{b-a}, & a \leq x < b \\ 1 & , x \geq b \end{cases}$$

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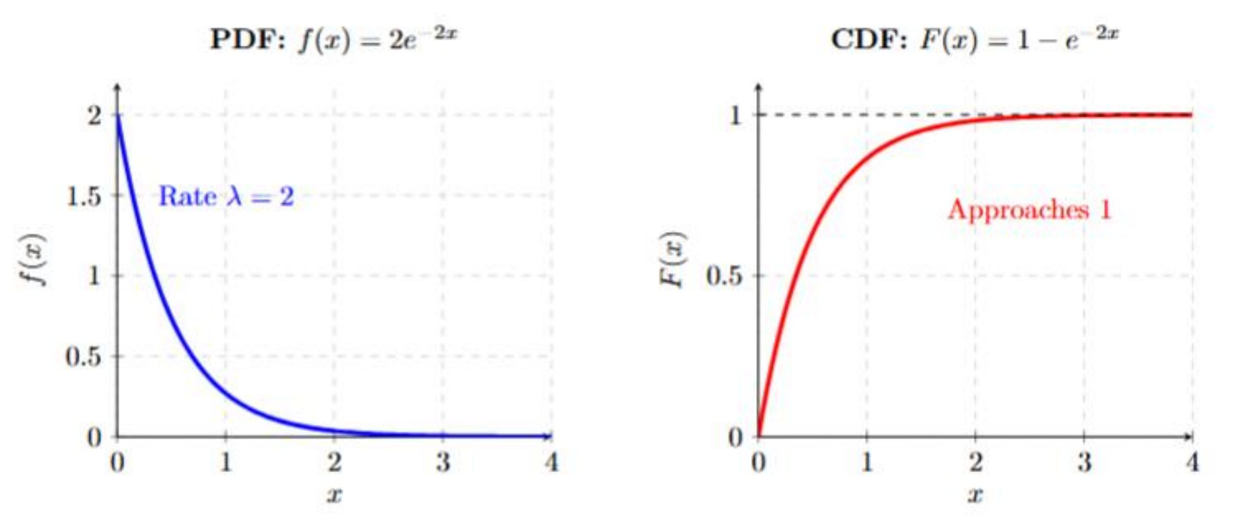

Exponential distribution $X \sim Expon(\mu = 2)$

$$f(x) = \begin{cases} 0\,, & x < 0 \\ \mu e^{-\mu x} & , x \geq 0 \end{cases} \qquad F(x) = \begin{cases} 0\,, & x < 0 \\ 1 - e^{-\mu x} & , x \geq 0 \end{cases}$$

Practice exercises/examples

(may not be covered during lectures but they are offered for your practice)

Exercise 1

- Consider the CRV has PDF $f(x) = \frac{1}{\pi(1+x^2)}$ (it is called Cauchy distribution)

- Find the PDF of the CRV: $Y = \frac{1}{X}$